

Prediction-Aware Robust Cross-Layer Rate-Splitting Multiple Access for Dynamic Multi-User Underwater Optical Wireless Communication

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Abstract—Underwater optical wireless communication (UOWC) provides high data-rate communication over short-range underwater links, but its performance is strongly affected by absorption, scattering, turbulence, pointing errors, and temporal channel variations. Existing UOWC transmission schemes mainly adapt optical parameters such as transmit power, beam divergence, and modulation according to the available channel condition. On the other hand, existing RSMA-based UOWC approaches mainly focus on precoding and rate or power allocation using the available channel information. However, these approaches generally do not jointly exploit future channel information, prediction uncertainty, and the interaction between optical transmission parameters and multi-user RSMA resources. In rapidly varying underwater environments, optimizing transmission only from the current or predicted channel can therefore result in unreliable resource allocation.

To address this gap, this paper proposes a prediction-aware robust cross-layer RSMA framework for dynamic multi-user UOWC. First, pilot-based channel estimation is employed to obtain the current channel state, and an autoregressive (AR) model is used to predict the channel for the next transmission interval. The prediction error is then quantified and incorporated into the transmission design through a reliability and uncertainty model. Based on the predicted channel and its uncertainty, the proposed framework jointly adapts the optical-layer parameters, including beam divergence, transmit power, modulation order, and data rate, together with RSMA common and private power allocation and precoding. A robust optimization problem is formulated to maximize the minimum achievable user rate while satisfying transmit-power, bit-error-rate (BER), and outage constraints. Monte Carlo simulations are used to evaluate the proposed framework in terms of channel prediction accuracy, achievable rate, sum rate, minimum-user rate, BER, outage probability, fairness, and energy efficiency. The proposed framework is designed to provide proactive and reliable resource allocation by exploiting future channel information while reducing the impact of channel prediction uncertainty in dynamic underwater environments.

Index Terms—Underwater optical wireless communication, rate-splitting multiple access, channel prediction, robust optimization, adaptive transmission, turbulence, pointing errors.

I. INTRODUCTION

Underwater wireless communication is important for applications such as environmental monitoring, ocean observation,

underwater sensor networks, autonomous underwater vehicles (AUVs), remotely operated vehicles, and the Internet of Underwater Things (IoUT). Although acoustic communication is widely used for underwater links, its limited bandwidth and relatively large propagation delay can restrict its suitability for high-data-rate applications. Underwater optical wireless communication (UOWC) has therefore attracted significant attention because of its potential to provide high-speed communication over short and moderate underwater distances. However, optical propagation in water is strongly affected by absorption and scattering, while turbulence causes random fluctuations in the received optical intensity. In addition, pointing errors caused by relative transmitter–receiver movement and imperfect alignment can further reduce the received optical power. These effects make reliable high-rate UOWC transmission challenging in dynamic underwater environments.

Several approaches have been investigated to improve the adaptability of UOWC systems, including adaptive beam divergence, transmit-power control, adaptive modulation, channel estimation, and beam steering. These methods show that a fixed transmission configuration is not suitable for all underwater channel conditions. However, most existing adaptive UOWC approaches primarily optimize individual optical parameters and do not jointly consider the allocation of resources among multiple users. Moreover, when the underwater channel varies with time, a transmission configuration optimized using the current channel condition may become unsuitable during the subsequent transmission interval.

For multi-user communication, non-orthogonal multiple access (NOMA) and rate-splitting multiple access (RSMA) provide mechanisms for allowing multiple users to share the same communication resource. NOMA relies on successive interference cancellation and user ordering, whereas RSMA provides greater flexibility by dividing the transmitted information into common and private streams. The common stream can be decoded by multiple users, while the private streams are decoded individually after common stream decoding. Recent RSMA-based UOWC studies have demonstrated the potential of RSMA for improving resource utilization through precod-

ing, power allocation, and rate allocation. However, these approaches mainly consider the available channel information and do not explicitly exploit future channel conditions or the uncertainty associated with channel prediction.

The above limitation becomes important in dynamic underwater environments. A channel estimated at the current time may differ significantly from the channel experienced during the next transmission interval. Channel prediction can provide information about this future state and enable proactive transmission adaptation. Nevertheless, a predicted channel is not an exact representation of the future channel because prediction errors arise from temporal channel variations, estimation errors, and model limitations. Therefore, directly treating the predicted channel as perfect channel state information can lead to an unreliable resource-allocation decision. This creates a need for a transmission framework that can exploit future channel information while explicitly accounting for prediction uncertainty.

In this work, a prediction-aware robust cross-layer RSMA framework is proposed for dynamic multi-user UOWC. Pilot-based channel estimation is first used to obtain the available channel information, and an autoregressive (AR) model is then employed to predict the channel for the next transmission interval. The prediction error is quantified and used to obtain a channel reliability measure. The predicted channel and its associated uncertainty are subsequently provided to a joint optimization stage. At the optical layer, beam divergence, transmit power, modulation order, and data rate are adapted, while at the RSMA layer, common-stream power, private-stream powers, common-rate allocation, and precoding are jointly selected. The proposed formulation aims to maximize the robust minimum-user throughput while satisfying transmit-power, bit-error-rate (BER), and outage constraints.

The main contributions of this work are summarized as follows:

- A dynamic two-user UOWC-RSMA system model is developed by jointly considering underwater attenuation, turbulence, pointing errors, and temporal channel variation.
- A pilot-based channel estimation and AR-based channel prediction mechanism is developed to obtain future channel information, while the prediction error is used to quantify the reliability of the predicted channel.
- A prediction-aware robust cross-layer transmission framework is proposed in which optical transmission parameters and RSMA resources are jointly adapted according to the predicted channel and its uncertainty.
- A robust minimum-user-rate optimization problem is formulated subject to transmit-power, BER, and outage constraints. The proposed framework is evaluated against conventional transmission schemes using rate, BER, outage probability, fairness, and energy-efficiency metrics.

The remainder of this paper is organized as follows. Section II presents the system and underwater optical channel model. Section III describes the pilot-based channel estimation and AR-based channel prediction. Section IV presents the

proposed robust cross-layer RSMA optimization framework. Section V presents the simulation results and discussion. Finally, Section VI concludes the paper.

II. SYSTEM AND CHANNEL MODEL

A. Multi-User UOWC System

Consider a two-user underwater optical wireless communication (UOWC) system in which an optical transmitter simultaneously serves two users, denoted by User 1 and User 2, using rate-splitting multiple access (RSMA). The two users share the same optical communication resource. The underwater channel is time-varying due to absorption and scattering, turbulence-induced fading, and transmitter–receiver pointing errors.

Let $h_k[n]$ denote the channel coefficient between the transmitter and User k at channel sample n , where $k \in \{1, 2\}$. The overall channel coefficient is modeled as

$$h_k[n] = h_{\text{att},k} h_{\text{turb},k}[n] h_{\text{PE},k}[n]. \quad (1)$$

Here, $h_{\text{att},k}$ represents deterministic propagation attenuation, $h_{\text{turb},k}[n]$ represents the time-varying turbulence component, and $h_{\text{PE},k}[n]$ represents the pointing-error component. This channel model provides the basis for channel estimation, temporal prediction, and adaptive RSMA resource allocation.

B. Underwater Optical Channel

The total attenuation coefficient of the underwater optical channel is given by

$$c = a + b, \quad (2)$$

where a and b denote the absorption and scattering coefficients, respectively. According to the Beer–Lambert law, the deterministic attenuation for User k is

$$h_{\text{att},k} = \exp(-cd_k), \quad (3)$$

where d_k is the propagation distance between the transmitter and User k .

For the considered two-user system, the distances are selected as

$$d_1 = 10 \text{ m}, \quad d_2 = 15 \text{ m}. \quad (4)$$

The absorption and scattering coefficients are selected as

$$a = 0.05, \quad b = 0.15, \quad (5)$$

which gives

$$c = 0.20. \quad (6)$$

Therefore, the two users experience different deterministic attenuation levels because of their different propagation distances.

C. Turbulence Model

Underwater turbulence produces random fluctuations in the received optical intensity. To represent the temporal variation of the channel, a correlated stochastic process is used. Let $x_k[n]$ denote the underlying temporal process for User k . It is generated using a first-order autoregressive process as

$$x_k[n] = \rho x_k[n-1] + \sqrt{1-\rho^2} w_k[n], \quad (7)$$

where $w_k[n]$ is a zero-mean, unit-variance Gaussian random variable and ρ is the temporal correlation coefficient.

The turbulence-induced channel coefficient is modeled using a log-normal representation as

$$h_{\text{turb},k}[n] = \exp\left(\sqrt{\sigma_I^2} x_k[n] - \frac{\sigma_I^2}{2}\right), \quad (8)$$

where σ_I^2 represents the turbulence-induced intensity fluctuation parameter. In the simulations,

$$\sigma_I^2 = 0.3, \quad \rho = 0.98. \quad (9)$$

The high value of ρ represents a slowly varying underwater channel and provides temporal information that can be exploited for channel prediction.

D. Pointing Error Model

Pointing errors occur because of relative transmitter-receiver movement and imperfect optical alignment. The effective beam radius at the receiver plane is modeled as

$$w_k = w_0 + d_k \tan\left(\frac{\theta}{2}\right), \quad (10)$$

where w_0 is the initial beam radius and θ is the beam divergence angle.

The horizontal and vertical pointing displacements are modeled as independent Gaussian random variables,

$$\Delta x_k[n] \sim \mathcal{N}(0, \sigma_p^2), \quad \Delta y_k[n] \sim \mathcal{N}(0, \sigma_p^2). \quad (11)$$

The corresponding radial displacement is

$$r_k[n] = \sqrt{\Delta x_k^2[n] + \Delta y_k^2[n]}. \quad (12)$$

The pointing-error channel coefficient is then given by

$$h_{\text{PE},k}[n] = \exp\left(-\frac{2r_k^2[n]}{w_k^2}\right). \quad (13)$$

The parameters used in the simulations are

$$w_0 = 0.005 \text{ m}, \quad \theta = 10^{-3} \text{ rad}, \quad \sigma_p = 0.002 \text{ m}. \quad (14)$$

The resulting channel in (1) therefore captures both deterministic propagation loss and time-varying random channel effects.

III. CHANNEL ESTIMATION AND TEMPORAL PREDICTION

Pilot-based channel estimation is first employed to obtain the current channel state. The temporal correlation of the estimated channel is then exploited to predict the channel for the next transmission interval.

A. Pilot-Based Channel Estimation

For a known pilot symbol s_p transmitted with power P_p , the received pilot signal at User k is

$$y_{p,k}[n] = \sqrt{P_p} h_k[n] s_p + n_k[n]. \quad (15)$$

The least-squares channel estimate is therefore

$$\hat{h}_k[n] = \frac{y_{p,k}[n]}{\sqrt{P_p} s_p}, \quad \hat{g}_k[n] = |\hat{h}_k[n]|. \quad (16)$$

The simulation uses $s_p = 1$, $P_p = 1$, and $N_{\text{est}} = 0.001$.

B. AR-Based Channel Prediction

To obtain a suitable representation for temporal prediction, the estimated channel magnitude is transformed into the logarithmic domain:

$$z_k[n] = \ln(\hat{g}_k[n] + \epsilon), \quad (17)$$

where ϵ is a small positive constant. The available $N = 10000$ samples are divided into 70% training and 30% testing data.

An AR model of order $p = 10$ is used to predict the next channel state:

$$z_k[n] = \sum_{i=1}^p \phi_{k,i} z_k[n-i] + e_k[n]. \quad (18)$$

The AR coefficients are obtained from the training data using least-squares estimation,

$$\phi_k = \mathbf{X}_k^\dagger \mathbf{Y}_k. \quad (19)$$

The one-step-ahead prediction is

$$\hat{z}_k[n+1] = \phi_k^T [z_k[n], \dots, z_k[n-p+1]]^T, \quad \hat{g}_k[n+1] = e^{\hat{z}_k[n+1]}. \quad (20)$$

C. Prediction Error and Reliability

The prediction error and NMSE are defined as

$$e_k[n] = g_k[n] - \hat{g}_k[n], \quad \text{NMSE}_k = \frac{\sum e_k^2[n]}{\sum g_k^2[n]}. \quad (21)$$

The prediction reliability is represented by

$$R_k = \frac{1}{1 + \text{NMSE}_k}, \quad G_k = \hat{g}_k^2 R_k, \quad (22)$$

where G_k denotes the reliability-aware effective channel quality. Thus, both the predicted channel strength and prediction confidence are considered in the subsequent RSMA resource-allocation stage.

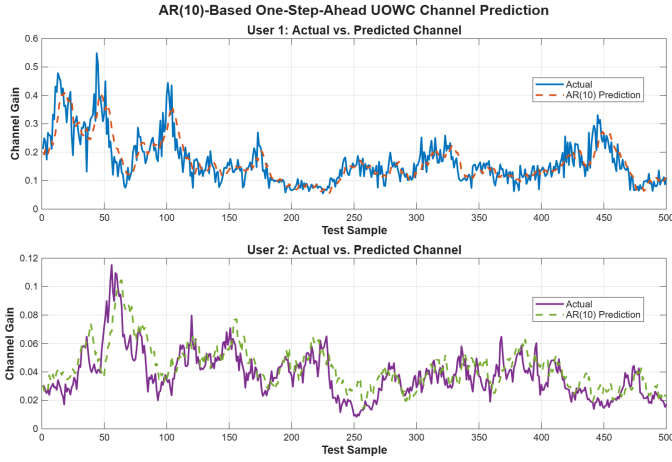


Fig. 1. Actual and AR(10)-predicted UOWC channel gain for the two users.

Fig. 2 shows that the AR(10) model captures the temporal variation of the UOWC channel for both users. The resulting prediction error is quantified using NMSE and incorporated into the prediction reliability used for subsequent robust resource allocation.

IV. PROPOSED ROBUST CROSS-LAYER RSMA OPTIMIZATION

The predicted channel provides information about the next transmission interval, but it is affected by prediction error. Therefore, the proposed framework jointly considers the predicted channel and its uncertainty when selecting the transmission configuration.

A. RSMA Transmission Model

The transmitted RSMA signal is expressed as

$$x = \sqrt{P_c}s_c + \sqrt{P_1}s_1 + \sqrt{P_2}s_2, \quad P_c + P_1 + P_2 \leq P_t, \quad (23)$$

where P_c is the common-stream power and P_1 and P_2 are the private-stream powers. The received signal at User k is

$$y_k = h_k x + n_k. \quad (24)$$

Each user first decodes the common stream and subsequently performs SIC to detect its private stream. The common-stream SINR and private-stream SINR are given by

$$\gamma_{c,k} = \frac{|h_k|^2 P_c}{|h_k|^2 (P_1 + P_2) + N_0}, \quad \gamma_{p,k} = \frac{|h_k|^2 P_k}{|h_k|^2 P_j + N_0}, \quad j \neq k \quad (25)$$

The achievable user rate is

$$R_k = R_{c,k} + R_{p,k}, \quad R_{c,k} = \log_2(1 + \gamma_{c,k}), \quad R_{p,k} = \log_2(1 + \gamma_{p,k}). \quad (26)$$

B. Robust Predicted Channel

Let $\hat{h}_k$ denote the predicted channel for the next transmission interval. The actual channel is modeled as

$$h_k = \hat{h}_k + e_k, \quad |e_k| \leq \epsilon_k, \quad (27)$$

where e_k represents the prediction uncertainty. The uncertainty bound is obtained from the prediction error.

The optimization variables are collected as

$$\Theta = \{\theta, P_t, P_c, P_1, P_2, M, r_d, R_{c,1}, R_{c,2}\}, \quad (28)$$

where θ , M , and r_d denote the beam divergence, modulation order, and transmission data rate, respectively.

C. Robust Cross-Layer Optimization

The proposed framework maximizes the reliable minimum-user throughput rather than only the instantaneous sum rate. The robust optimization problem is formulated as

$$\max_{\Theta} \min_{k \in \{1,2\}} \min_{h_k \in \mathcal{H}_k} R_k(h_k, \Theta) \quad (29)$$

subject to

$$P_c + P_1 + P_2 \leq P_{\max}, \quad \text{BER}_k \leq \text{BER}_{\max}, \quad P_{\text{out},k} \leq P_{\text{out}}^{\max}. \quad (30)$$

The feasible transmission configuration includes optical-layer parameters and RSMA resources. For each predicted channel state, candidate configurations are evaluated under the uncertainty bound. Configurations violating the power, BER, or outage constraints are discarded, and the feasible configuration providing the largest minimum-user rate is selected.

The resulting operation is

$$\text{Prediction} \rightarrow \text{Uncertainty} \rightarrow \text{Joint Optimization} \rightarrow \text{RSMA Transmission}. \quad (31)$$

V. SIMULATION RESULTS AND DISCUSSION

The proposed prediction-aware robust RSMA framework is evaluated for a two-user dynamic UOWC system using Monte Carlo simulation. The considered link distances are 10 m and 15 m, with absorption coefficient $a = 0.05$, scattering coefficient $b = 0.15$, turbulence variance $\sigma_I^2 = 0.30$, and temporal correlation $\rho = 0.98$. A total of 10^4 channel samples are generated, of which 7000 samples are used for training and 3000 for testing. The AR prediction order is set to 10. The maximum normalized transmit power is one, while the BER and outage constraints are 10^{-2} and 0.1, respectively. The target rate for outage evaluation is 0.20 bits/s/Hz.

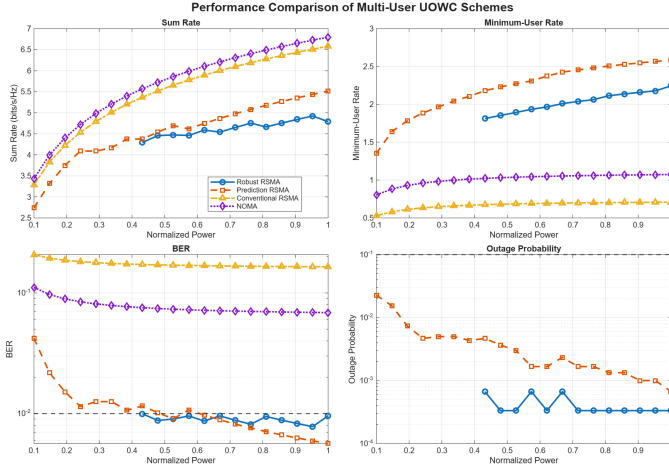


Fig. 2. Actual and AR(10)-predicted UOWC channel gains.

A. Channel Prediction Performance

Fig. 2 compares the actual and AR(10)-predicted channel gains for the two users. The predicted channels follow the general temporal variations of the actual channels, although rapid channel fluctuations are not reproduced exactly. The obtained NMSE values are 0.0514 and 0.0670 for Users 1 and 2, respectively. The corresponding prediction reliability values are 0.9511 and 0.9372. These results indicate that the predicted channel provides useful future channel information while retaining a measurable prediction uncertainty.

B. Rate Performance

Fig. ?? compares the proposed robust RSMA with prediction-based RSMA, conventional RSMA, and NOMA. The sum-rate performance increases with normalized transmit power for all schemes. NOMA provides the highest aggregate rate, followed by conventional RSMA, prediction-based RSMA, and the proposed robust RSMA. This behavior is expected because the proposed method prioritizes reliable operation under channel-prediction uncertainty rather than maximizing the instantaneous sum rate.

The minimum-user rate also increases with transmit power. The proposed robust scheme becomes feasible at a normalized transmit power of about 0.432, since no configuration below this level simultaneously satisfies the imposed BER and outage constraints. At the maximum normalized power, the proposed scheme achieves a minimum-user rate of approximately 2.24 bits/s/Hz and a sum rate of approximately 4.92 bits/s/Hz. The lower aggregate rate compared with non-robust schemes represents the throughput cost of explicitly incorporating prediction uncertainty and reliability constraints.

C. BER and Outage Performance

The BER and outage results are shown in Fig. ?. The proposed robust RSMA maintains the BER close to or below the prescribed 10^{-2} threshold over its feasible operating region. More importantly, its outage probability remains below the specified 0.1 constraint, demonstrating the effect of robust

configuration selection. The prediction-based scheme without robustness provides higher rate but exhibits a larger outage probability because the prediction uncertainty is not explicitly included in its transmission design.

Overall, the results demonstrate the principal trade-off of the proposed framework: robustness and reliability are obtained at the cost of some aggregate spectral efficiency. The proposed method therefore provides a proactive transmission strategy in which future channel information is used while prediction uncertainty is explicitly considered during resource allocation.

VI. CONCLUSION

This paper presented a prediction-aware robust cross-layer RSMA framework for dynamic multi-user underwater optical wireless communication. The proposed approach combines pilot-based channel estimation, AR(10) channel prediction, prediction-error-based reliability estimation, and robust RSMA resource allocation. Unlike conventional approaches that rely mainly on the current channel condition, the proposed framework uses the predicted future channel while explicitly considering prediction uncertainty.

The simulation results demonstrate that the AR-based predictor can capture the temporal behavior of the underwater optical channel, with NMSE values of 0.0514 and 0.0670 for Users 1 and 2, respectively. The corresponding prediction reliability values are 0.9511 and 0.9372. The robust transmission strategy becomes feasible at a normalized transmit power of approximately 0.432 under the considered BER and outage constraints. At the maximum normalized transmit power, the proposed scheme achieves a robust minimum-user rate of approximately 2.24 bits/s/Hz and a sum rate of approximately 4.92 bits/s/Hz.

The results also highlight the throughput-reliability trade-off of robust transmission. Although conventional schemes can provide higher aggregate throughput, the proposed method explicitly accounts for prediction uncertainty and reliability constraints, making it suitable for dynamic underwater environments where the future channel cannot be known perfectly. Future work will consider more advanced prediction models, larger numbers of users, multi-transmitter/multi-element optical configurations, and more efficient optimization methods for real-time implementation.